



Day 26: DNS and Route 53 for Domain Management

[General Interview Questions]

◆ 1. What is a public hosted zone in Route 53?

Answer:

It's a DNS zone that makes **DNS records publicly available on the Internet** for a registered domain.

◆ 2. How do you map a domain name to an EC2 instance using Route 53?

Answer:

Create a **public hosted zone**, add an **A record**, and point it to the EC2 instance's **public IPv4 address**.

◆ 3. What challenge occurs when mapping to EC2 public IP directly?

Answer:

If the instance is **stopped or restarted**, it gets a new public IP, **breaking the A record** unless **Elastic IP** is used.

◆ 4. How does an Alias record differ from a standard A record?

Answer:

Alias records can point to **AWS resources** (e.g., S3, ELB) and **don't cost for queries**. They are AWS-specific.

◆ 5. What are the different routing policies available in Route 53?

Answer:

- Simple
- Weighted
- Latency-based
- Failover

- **Geolocation**
- **Geoproximity** (via Route 53 Traffic Flow)

◆ 6. When would you use weighted routing?

Answer:

When you want to **split traffic** between multiple resources (e.g., 70% to US server, 30% to EU).

◆ 7. What is a failover routing policy?

Answer:

Used for **active-passive setups**. If the primary fails (health check fails), traffic is routed to the secondary.

◆ 8. How do you create a latency-based routing setup?

Answer:

Deploy resources in **multiple regions**, then use **latency policy** to direct users to the **lowest-latency region**.

◆ 9. What is the importance of TTL in DNS records?

Answer:

TTL controls how long DNS results are **cached**. Lower TTL = faster propagation, higher TTL = reduced lookup cost.

◆ 10. Can you route traffic differently for users in different countries?

Answer:

Yes, using **Geolocation routing policy**. You can route based on the **country or continent** of the user.

◆ 11. What happens if a region in a latency policy becomes unavailable?

Answer:

Without a failover or health check, Route 53 may still try routing there unless **failover or health-based rules** are in place.

◆ 12. What's the best method to simulate failover in Route 53?

Answer:

Set up a **failover policy**, attach **health checks**, then simulate failure (e.g., stop EC2) and monitor failover to secondary.

◆ **13. Can you use Route 53 to manage private DNS for VPCs?**

Answer:

Yes. Create a **private hosted zone**, associate it with a **VPC**, and create records only accessible within that VPC.

◆ **14. What role does health check play in routing policies?**

Answer:

Health checks are essential in **failover and latency-based** routing to ensure traffic is directed to **healthy endpoints**.

◆ **15. How can you manage blue/green deployments with Route 53?**

Answer:

Use **weighted routing** to gradually shift traffic between versions or **failover** to switch instantly based on health.